

2026.8.27 早晨电路学习感悟与笔记

- 1.区分电桥整体的输出与内部的响应表现
- 2.戴维南等效：开路电压 等效电阻 诺顿等效：短路电流
- 3.注意符号与正负方向，有时 I R U 真的会带负号

T1 (作业补充 | 最大功率)

开路电压: $24 = 24 + 4i = 3I + 8i + 2i$

必须使用黑色字迹书写: 在答题卡区域内作答, 超出黑色矩形边框限定区域的答案无效

$$24 = 3I + 6(2 - i)$$

$$8 = I + 2i$$

$$8 = 3I - 2i$$

$$I = 4A \quad i = 2A$$

$$U_a = 24 - 3 \times 4 - 8 \times 2 = -4V$$

$$U_b = -4V + 4V + 8V = 8V$$

$$U_b = -4V + 4V + 8V = 8V$$

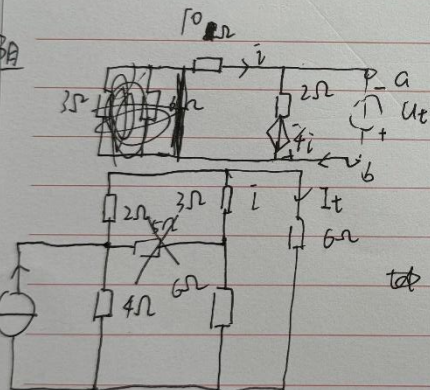
$$\therefore U_{oc} = U_b - U_a = 4V$$

$$2.5 \Omega$$

$$R_L = 4 \Omega$$

$$P_{max} = \frac{U_{oc}^2}{4R_{eq}} = 1.6W$$

等效电阻



$$U_t = 10i$$

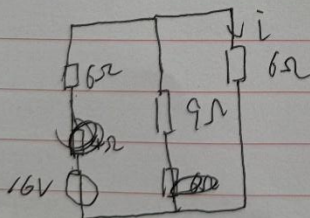
$$U_t = 4i + 2(4i - i)$$

$$= 2i + 2I_t$$

$$I_t = 4i$$

$$R_{eq} = 2.5 \Omega$$

拿下!



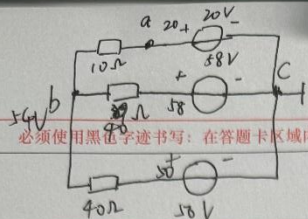
$$\frac{3.6}{3.6+6} \times 16 \times \frac{9}{9+6} \div 6 = 6 \times \frac{15}{15} = 3.6A$$

$$= 3.6A$$

$$\frac{16}{3.2} \times \frac{3}{5}$$

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3.



等效电阻 & 开路电压

30/34

20Ω

34V

17/15 A

必须使用黑色字迹书写：在答题卡区域内作答，超出黑色矩形边框限定区域的答案无效

$$50 - 40I_1 = 58 - 40I_2$$

对！
符号也ok！

$$I_1 = -I_2$$

$$I_2 = 0.1A$$

4.

$$6 \times (-1) + 10 \times \left(\frac{U_2}{-4}\right)$$

$$= (-2) \times 2 + (-3) \times U_2$$

$$6 + 2.5U_2 = 4 + 3U_2$$

9/10:4

3.2

叠加作业题

$$U_2 = 4V$$

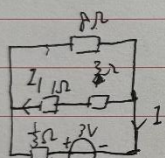
$$\frac{1}{3}A \quad -\frac{2}{3}A$$

$$I_L = I' + I''$$

② 电压源短路

电桥

① 电流表断路



$$I_L = I' = 1A$$

$$I_L = I = 0$$

$$R_{eq} = \frac{1}{3} + \frac{2}{3} = 3\Omega$$

$$I_L = I' = -\frac{1}{3}A$$

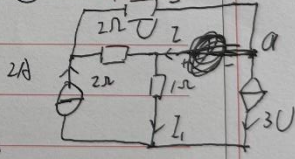
$$P = I_L^2 R = \frac{4}{9}W$$

$$I' = 1A$$

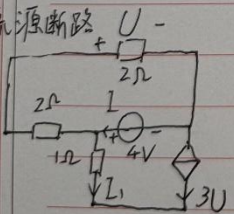
$$I'' = -\frac{2}{3}A$$

$$I_L = -6A$$

② 电压源短路



① 电流源断路



$$I - I_1 = I + 3U$$

$$I' = 5A$$

$$I_1 = -3U$$

$$U = 2V \quad 2(2 + I - I_1) + 2(I - I_1) = 0$$

$$4 = U + 2(I - I_1)$$

$$I_1 - I = 1$$

$$U = 2V$$

$$I - I_1 = 1$$

$$2 + I - I_1 = I + 3U$$

$$I_1 = I' + I''$$

$$I = I' + I''$$

$$= \frac{14}{3}A$$

$$= -\frac{14}{3}A$$

$$I_1 = I + 1 = 2 - 2I$$

$$I'' = \frac{1}{3}A$$

第1页

$$I_1'' = \frac{4}{3}A$$

$$2 - I_1 = 3U$$

$$6 - 3I_1 = 4 + 2I - 2I_1$$

$$I = AI_s + B$$

$$I = \frac{1}{2}I_s - 2$$

$$-1 = 2A + B$$

$$0 = 4A + B$$

$$A = \frac{1}{2}, B = -2$$

$$I_s = 6$$

$$52W$$

$$78W$$

$$U = AR + B$$

$$U = 25V$$

$$15 = 10A + B$$

$$20 = 20A + B$$

$$A = 0.5, B = 10$$

$$\frac{100}{36}$$

$$3-10$$

$$R_{eq} = 6 + \frac{24}{11}$$

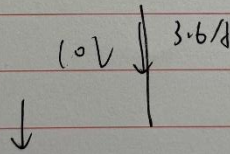
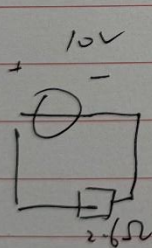
$$= \frac{90}{11}$$

$$\frac{3}{11} \times \frac{1.1}{\frac{49}{11}} \times \frac{1.1}{\frac{49}{11}}$$

$$\frac{49}{11}$$

$$9 \times \frac{49}{11}$$

$$2.4V$$



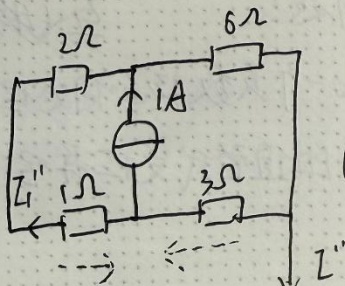
$$\frac{10.6V}{3.6A}$$

批改听一下音!

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1. 含电桥且公共边为源、

2. 有一些电压可直接用
欧姆定律



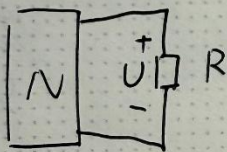
(1) 桥整体对外输出为 0

(2) 桥内部仍有电流

$$I'' = -\frac{3}{1+3} \times (1A) = -\frac{3}{4}A$$

戴维南等效

3.

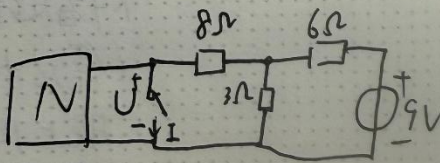


$$R=10 \quad U=15$$

$$R=\infty \quad U=20$$

$$U = \frac{U_{OC}}{R_i + R} \times R$$

4.



右侧先进行戴维南等效

左边也一样

(最后看一组等效题放松一下)